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Yijun Zhang and Han Bao contributed equally to this work.

Key Points:

- Regionalized patterns of spatial bias in both Commander Island and Tohoku regions suggest the effects of 3D Earth structure on BP Inferred locations
- We proposed a regional SEBP method to account for spatially variable slowness correction, which has reduced the location uncertainty by 50%
- The enhanced dynamic weakening mechanism may not be activated during the 2011 Tohoku earthquake

Supporting Information:

Supporting Information may be found in the online version of this article.

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Understanding and Mitigating the Spatial Bias of Earthquake Source Imaging With Regional Slowness Enhanced Back-Projection

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Abstract

We investigate characteristics of spatial biases in the Back-projection method and the effectiveness of the Slowness Enhanced Back-projection (SEBP). The spatial bias refers to the location error in Back-projection caused by travel time errors due to 3D Earth structure. SEBP reduces such bias by calibrating the gradient of travel time (slowness) in the source. We first analyze 22 M4–M6 earthquakes in the Commander Island and find that the amplitudes and directions of spatial biases follow distinct regional patterns. In light of this, we propose a regional SEBP approach that introduces spatially variable slowness correction. The regional SEBP in the Commander Island resulted in a ∼50% reduction in the average length of spatial bias from ∼20 to ∼10 km, which is more effective than the uniform SEBP that gives a 25% error reduction. We then analyze 109 M4–M7 earthquakes in the Tohoku region and also find a 50% error reduction by regional SEBP. This indicates that half of the spatial biases are aleatory uncertainties that are caused by regional structural complexity and can be calibrated with SEBP, while the residue errors are epistemic uncertainties that are random and caused by local velocity heterogeneities. With regional SEBP applied to the 2011 Tohoku earthquake, we find that high-frequency radiators did not reach beyond the down-dip limit of interplate seismicity, indicating that the coseismic slip unlikely penetrated the down-dip velocity-strengthening zone. Such observations suggest that the enhanced dynamic weakening mechanism due to thermal pressurization effects may not be activated during this event.

Plain Language Summary

Back-projection (BP) is an array processing technique that takes advantage of coherent earthquake signals to track the source of energy. It relies on accurate source-receiver travel times to determine the source locations. However, the difference between the assumed 1D velocity model and the real 3D velocity structure causes location uncertainties in BP results. We systematically study different sources of back-projection uncertainties (performance of waveform alignment, earthquake size, and BP's parameterization) due to 3D Earth structure and quantitatively evaluate their effects on location errors of BPs. Using the Commander Island as a test area, we find regionalized patterns of location errors corresponding to different fault segments, suggesting that variable slowness correction is needed for the improvement in location accuracy. Motivated by this, here we propose a regional slowness enhanced method to calibrate the 3D velocity structure effect using aftershock data. We validate this method by applying it to both the Commander Island and the Tohoku regions, respectively, and it reduces location errors by half in both regions. Based on this proposed method, the calibrated rupture details of the 2011 Tohoku earthquake provide new evidence of where its rupture stopped, shedding new insights into the rupture mechanisms of large earthquakes.

1. Introduction

Back-projection (BP) is an observational technique that tracks the growth of large earthquake ruptures from coherent seismic phases recorded across a seismic array (see Kiser & Ishii, 2017, for a comprehensive review of BP). High-frequency rupture characteristics resolved from BP provide unique perspectives on understanding detailed rupture kinematics and geometrical complexities of large earthquakes (Bao et al., 2019; Fukahata et al., 2014; Ishii et al., 2005; Kiser & Ishii, 2011; Meng, Ampuero, Luo, et al., 2012, 2012b; Meng et al., 2018, 2019; Okuwaki et al., 2014; Okuwaki & Yagi, 2018; Satriano et al., 2012; Wang & Mori, 2011a; b). Recent studies also demonstrated the effectiveness of the BP method in detecting and locating seismicity (Fan & Shearer, 2016; Feng et al., 2020; Kiser & Ishii, 2013).

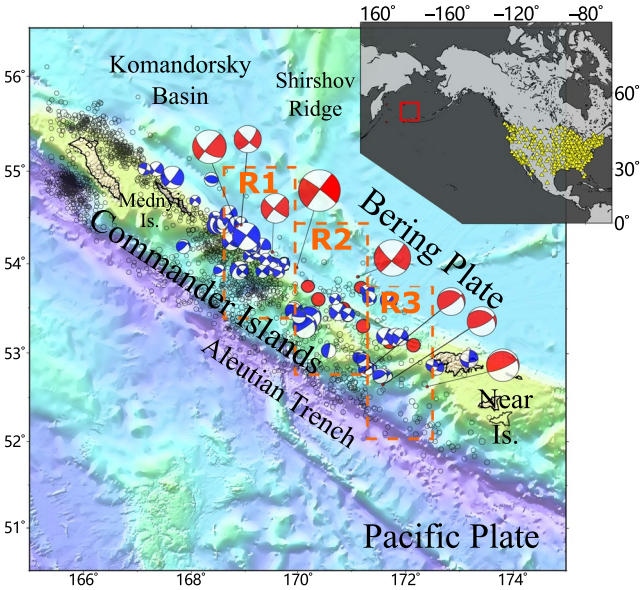

Figure 1. Twenty-two M4-6 earthquakes within the Commander Islander region and the seismic stations used. The 22 earthquakes (red circles and beach balls) that occurred from 2013 to 2017 are all shallow events (epicenter depth <35 km based on ISC catalog, http://www.isc.ac.uk). GCMT Focal mechanisms of eight events (red beach balls) and Mw > 5 earthquakes (blue beach balls) from 1976 to 2017 are shown (Ekström et al., 2012). The top-right inset map shows the plotted source region (red box) and the US seismic network (yellow triangles). The three dashed rectangular boxes are the three zoned regions that are selected based on the tectonic characteristics of this region (Lutikov et al., 2019; Newberry et al., 1986; Ruppert et al., 2012).

However, previous studies found that locating seismicity using BP suffers from notable spatial bias, *De*, defined as the vector deviation between an event's BP inferred (BPI) location from its catalog location. Events' catalog locations are usually relocated with local or regional networks and are thus considered to be more accurate than BPI locations. The uncertainty of BPI location is attributed to both the noise and travel time errors. Since the ambient noise is weak compared with the waveform amplitude of $M > 6$ earthquakes imaged with BP, the noise relevant to BP refers to the incoherent part of the waveforms caused by coda waves, local scattering, and variation of site effects. Meng et al. (2012a, 2012b) show that the perturbation of BPI location due to noise is on the order of 1–4 km depending on the noise amplitude and the array aperture. A bigger source of *De* is the travel time errors introduced by approximating the 3D Earth velocity structure with a 1D reference model. Previous teleseismic BP studies in Tohoku, Nepal, Mexico, and Chile found that the spatial bias associated with travel-time errors is on the order of 20–30 km (Fan & Shearer, 2017; Meng et al., 2016, 2019; Palo et al., 2014; Zeng et al., 2020). In addition to BP spatial bias due to travel time errors, waveform complexity such as water reverberation and coda waves generated by near-trench 3D structures also leads to artificial sources in BP imaging (Fan & Shearer, 2016; Ihmlé & Madariaga, 1996; Wu et al., 2019; Yue et al., 2017). The structural effect also distorts the BP amplitude. Okuwaki et al. (2019) show that the amplitude variation of Green's functions results in the depth-dependent intensity of BP sources along a subducting plate interface.

Understanding spatial biases of BP is of significance not only for detecting and locating epicenters of earthquakes but also for observational studies of large earthquakes. This is because spatial biases can distort measurements of rupture properties (e.g., rupture lengths, speeds, and directivities), and thus result in erroneous interpretations of earthquake physics (Bao et al., 2019). Since the spatial biases are primarily caused by our insufficient knowledge of the 3D velocity heterogeneity, several studies attempted to calibrate the travel times using aftershocks near the mainshock rupture (Ishii et al., 2007; Meng et al., 2016; Palo et al., 2014). For instance, Ishii et al. (2007) measured the time shifts to align the aftershock arrivals and obtained the time correction for a given source grid point from the weighted-average of the nearby aftershock measurements. Palo et al. (2014) further separated the travel time correction into two terms: a static term specific to a station and a dynamic term specific to each pair of source grid points and stations. The dynamic term is obtained by interpolating time shifts of aftershocks over the source area. The aftershock calibration typically requires moderate-size events (M4.5–6) which are small enough to be approximated as a point source in the far-field yet large enough with sufficient Signal-to-Noise ratio (SNR) at the teleseismic receivers. A practical limitation is that the sparsity and uneven distribution of such aftershocks may lead to unreliable correction if the travel times are interpolated empirically. Alternatively, Meng et al. (2016) developed a parametric approach named Slowness Enhanced Back-projection (SEBP). Instead of direct spatial interpolation of the travel times, a slowness correction term (spatial gradient of travel time) is derived from the aftershock measurements. The total travel-time correction for a given source grid point includes the static term and the vector product of the source location and the slowness term. This slowness term is assumed to be uniform for the whole rupture region, which provides a linear contribution to the correction of travel-time errors. It is a parametric method that can still obtain the travel-time correction over the whole region even at the sparsity of aftershocks. Although SEBP has been applied to several large earthquakes with rupture lengths less than 150 km (Bao et al., 2019; Meng et al., 2016, 2018, 2019; Sangha et al., 2017), the effectiveness of using a uniform slowness correction term has not been systematically investigated for earthquakes of larger sizes. Recently the availability of 3D velocity models from seismic tomography enables more accurate travel-time prediction, which provides a direct way to mitigate the spatial bias of BP (Liu et al. (2017); Yang et al. (2020)).

To investigate the pattern of spatial biases in large source regions, we studied the Commander Island region (Figure 1), which spans more than 400 km along the western Aleutian trench. We analyzed 22 M 4–6 earthquakes

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that occurred from 2013 to 2017. We found a systematic correlation between the amplitude of *De* and the event separation distance which justifies the condition to apply slowness correction. We also found that the amplitudes and directions of *De* follow distinct regional patterns corresponding to each fault segment, likely caused by variations of the regional velocity structures. Our results suggest that it is challenging to effectively calibrate travel time errors for the entire source region with a uniform slowness correction term. Therefore, we propose a regional Slowness Enhanced Back-projection (SEBP) approach that introduces spatially variable slowness corrections. By applying regional SEBP to the Commander Region, we found that the average of *De* was reduced by 50% (from ∼20 to ∼10 km), more effective than the 25% error reduction of uniform SEBP. We further applied regional SEBP to the source region of the 2011 Mw 9.1 Tohoku earthquake, one of the largest and most complicated earthquakes ever recorded in history. We investigated spatial biases of over a hundred M4–M7 events and used the K-means clustering algorithm (Forgy, 1965; Macqueen, 1967) to automatically divide the source regions. We found a similar 50% reduction in *De* after adopting regional slowness corrections. By applying regional SEBP to the Tohoku mainshock, we found that its coseismic slip did not penetrate the down-dip limit of interplate seismicity. This finding is consistent in the BPs of both the United States (US) array and the European (EU) array. In summary, our studies quantify and reduce the spatial biases of BP, which gives rise to more accurate interpretations of earthquake source physics.

2. Investigation of Spatial Biases in the Commander Island Region

2.1. Data and Method

To investigate characteristics of BP's spatial biases, we studied 22 moderate earthquakes (M4–M6) within the Commander Island region that occurred from 2013 to 2017 (Figure 1 and Table S1 in Supporting Information S1). We selected this region because of the abundance of moderate-size earthquakes (M4–M6) along the Bering Fault, the boundary between the Bering Plate and the Commander Island belt (Mets et al., 1990; Seliverstov, 2009). Moderate-size earthquakes have relatively high SNRs so that their source locations can be well resolved by BP. Since the source region is far from densely distributed seismic networks, the global catalog of the investigated earthquake locations may have large uncertainties. Because accurate event locations are crucial for the calibration of SEBP, we performed teleseismic double-difference relocation on the nearby earthquakes (Pesicek et al., 2010) (Figure S1, Table S2 in Supporting Information S1). Their relative location uncertainties after relocation drop to 1–3 km, sufficiently accurate for them as reference locations to calibrate BPI locations in the Commander Island. For the BP analysis, we adopted the waveforms of 336 stations in the continental US which provide a large range and azimuthal coverage (Figure 1). We select stations with epicentral distances between 30° and 90°, such that waveform complexity caused by mantle discontinuities and the core-mantle boundary is absent, and that the contaminations of the P_{diff} phase are not included.

Waveform data are downloaded from the Data Management Center (DMC) of the Incorporated Research Institutions for Seismology (IRIS) (https://ds.iris.edu/wilber3). Results presented in the main text are obtained from seismic data that are normalized and aligned on the first P arrivals and filtered to 0.5–2.0 Hz. Two other frequency bands (0.05–0.5 Hz and 1–4 Hz) are also used for comparative analysis which can be found in the supplementary material. We ensure that for a given event, the chosen stations are away from its nodal plane and share the same P-wave polarity. We adopt these quality control criteria to isolate the effect of travel time errors on BP spatial bias from station configuration.

We applied two BP methods: the classic beamforming BP (Ishii et al., 2005) and the MUSIC BP (Meng et al., 2011). Theoretical travel times are calculated from the layered IASP91 velocity model (Kennett & Engdahl, 1991), and the "reference window" strategy (Meng et al., 2012a, 2012b) is used to mitigate the "swimming" artifacts, a smearing artifact in images along the direction toward the arrays due to the trade-offs between the radiation origin times and source-receiver distances (Koper et al., 2012). Results presented in the main text are produced by MUSIC BP, and those of beamforming BP can be found in the supplementary material for comparison.

2.2. Obtaining the Spatial Bias

Since investigating BP's spatial uncertainty is the key focus of this study, we first provide a clear definition of the spatial bias of BP and how it is obtained. To account for the travel time variations due to 3D Earth structures, BP routinely adopts a timing correction derived from the "hypocenter alignment" (Ishii et al., 2005, 2007). The first

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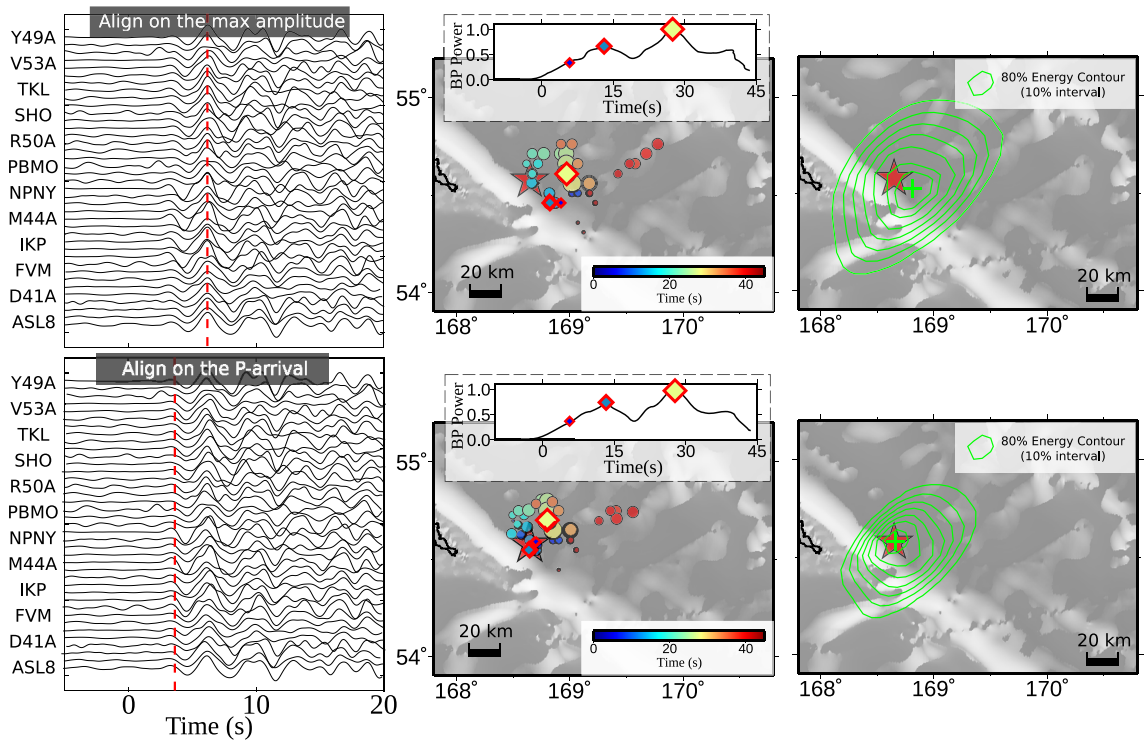

Figure 2. The nontriviality of hypocenter correction (waveform alignment) in Back-projection (BP) location. (left) Examples of velocity records (0.2–1.0 Hz) after hypocenter alignments whose 10-s alignment-window centers at the "maximum amplitude" or the "P-arrival". Waveforms are from an Mw 6.3 earthquake that occurred on 17 July 2017. The time "0" is the manually selected reference time for clearly visualizing the alignment of P wave arrivals. The dashed red lines indicate the aligned positions of the maximum amplitudes (top-left inset) and the initial arrivals (bottom-left inset) of P waves. (middle) High-frequency radiators (circles) of the Mw 6.3 events imaged by MUSIC BP are color-coded by time with sizes proportional to BP's beam power. The top insets illustrate BP's beam power as function of time. Three high-frequency radiators (diamond with the red edge) are picked for easy cross-reference. Radiators after 30 s appear to drift toward NEE, which is likely caused by depth phases of P waves. (right) Integrated BP results (0.5–2 Hz, 80% energy contour) of the first 10 s. The Back-projection inferred (BPI) locations are marked by green crosses.

arrival of the mainshock is assumed to come from the hypocenter. A set of travel time errors due to 3D structures is obtained by cross-correlating the initial P-waves. Such "hypocenter correction" is usually applied to the entire rupture region. However, since it is only accurate within close proximities to the hypocenter, it does not account for the variation of travel time errors at distant source locations. Here, we consider *De* as the additional location error after applying the "hypocenter correction". De would be zero only if ray paths from the hypocenter and distant sources sampled 3D earth structure in the same way.

To guarantee accurate estimations of *De*, we first ensure that the hypocenter alignment is adequately performed and has no influence on the measurement of *De*. After the waveforms are downloaded, they are first aligned based on travel times predicted by a 1D reference Earth velocity model (e.g., IASP91). We then adopt an iterative cross-correlation approach to align the waveforms of initial P-arrivals from low- to high-frequency bands. In the example shown in Figure 2, the waveforms are first aligned at 0.05–0.2 Hz in a 20-s window and then aligned at 0.2–1 Hz in a 10-s window. This procedure is repeated for 0.5–2 Hz if the SNR is higher than 0.5. To avoid cycle skipping issues, we limit the maximum allowed shift to the period of the upper bound of the frequency band. We further found that the position of the cross-correlating window is nontrivial. Although the peak amplitude is often more easily identified than the P-arrival, we note that the cross-correlating window should be centered at the P-arrival, not P-wave's peak amplitude, because the first arrival comes from the hypocenter of an earthquake. This is likely more important for M6+ events since their finite source effects are not negligible, and their centroid locations corresponding to the peak energy may have considerable deviations from their hypocenters. If the hypocenter alignment is performed correctly, a BP on the reference event itself (self-aligned BP) should collocate with its catalog location. Figure 2 shows the self-aligned BP of an Mw 6.3 earthquake when the cross-correlating window is centered at the maximum amplitude of waveforms versus that centered at the P-arrivals. In the former case, its BPI location (center of the integrated energy contours of the first 10-s BPs) is 10–13 km away from the

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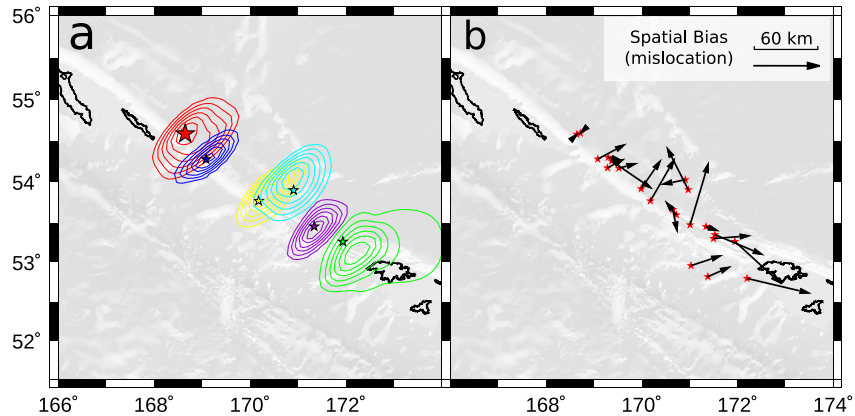

Figure 3. Overview of spatial biases in the Commander Island region: BP results with time corrections from the Mw 6.3 earthquakes as the reference event applied to the other 21 events. (a) Examples of 80% energy contour of integrated BP images of the first 10 s. For clarity, only 6 events are shown here. (b) The spatial biases (black vector) of the 21 events, as measured by the distance of the image peak from the epicenter (De), which is defined in the same way as Fan and Shearer (2017).

epicenter. In the latter case, such deviations are less than 1 km, which is negligible. The process of BP in determining locations of earthquakes or sub-sources with respect to the hypocenter is similar to that of the "master event" location technique (Ito, 1985). Once the hypocenter correction is obtained for a "reference event", it is applied to the waveforms of "investigated events" to obtain their BPI locations (relative to the reference event). Based on a similar argument of finite source effects, the data window corresponding to the BPI location should also be centered at the P-arrival, not the peak amplitude. In Fan and Shearer (2017), many of their self-aligned BP locations deviate from their catalog locations. This is likely caused by two factors. First, their cross-correlating window is not centered at P-arrivals. Second, they used a 20-s data window, longer than the durations of the investigated events. Therefore, the resulting BPI locations are more consistent with the centroid locations, not the hypocenter locations. Figure S2 in Supporting Information S1 shows the comparison when different window lengths are applied. We conclude that it is optimal to use a 10-s window centered at P-arrival for both the hypocenter alignment and obtaining BPI locations, demonstrated by the co-location between self-aligned BP and hypocenter location. Finally, we obtain the spatial bias by calculating the difference between the BPI location and the catalog location, similar to that of Fan and Shearer (2017). We here treat the relocated earthquake catalog as the ground truth because of the small errors of their relative locations (~1 km, Pesicek et al., 2010). Figure S3 in Supporting Information S1 gives an example of a M5.5 earthquake, the comparison of BP results when different BP parameters (stacking and integrating window lengths) are applied. We can see that various lengths of stacking and integrating windows in the BP method invariably located the kernel center of self-aligned BP images at its hypocenter, which is likely because a M5.5 earthquake can be regarded as a point source at teleseismic distances. It suggests that as the finite-fault effect of M5 earthquakes can be ignored at teleseismic distances, they can be well resolved despite the variation in BP parameters.

2.3. Regional Patterns of Spatial Biases in the Commander Island Region

Figure 3 shows an example of spatial biases in the Commander Island region, where the BP energy contours and *De* are plotted for an Mw 6.3 earthquake used as a reference event. We select 9 M5+ reference events with relatively high SNRs. In total, we analyze 22 events and 198 event pairs (Table S1 in Supporting Information S1).

We found that spatial biases (*De*) in Commander Island have an average amplitude of 21 km and manifest regional patterns in their directions and amplitudes. As shown in Figure 3b, *De* has distinct characteristics in the northwest, middle, and southeast parts of the study area. We segmented the investigated area into three sub-regions (Figure 1) and plotted the 198 measurements of *De* in the polar diagram (Figure 4). Figure 4 shows patterns of directivities and amplitudes of *De* when regions of reference events and investigated events are outlined. Measurements of *De* are plotted as bars in the polar plot. For convenience, we use the notation "R1→R2" to represent cases when reference events from Region 1 are used to estimate *De* of events in Region 2. In each polar plot, the mean value of *De* is marked (thick dashed line). In terms of the amplitude of *De*, we found that the mean value of *De* of R1→R1

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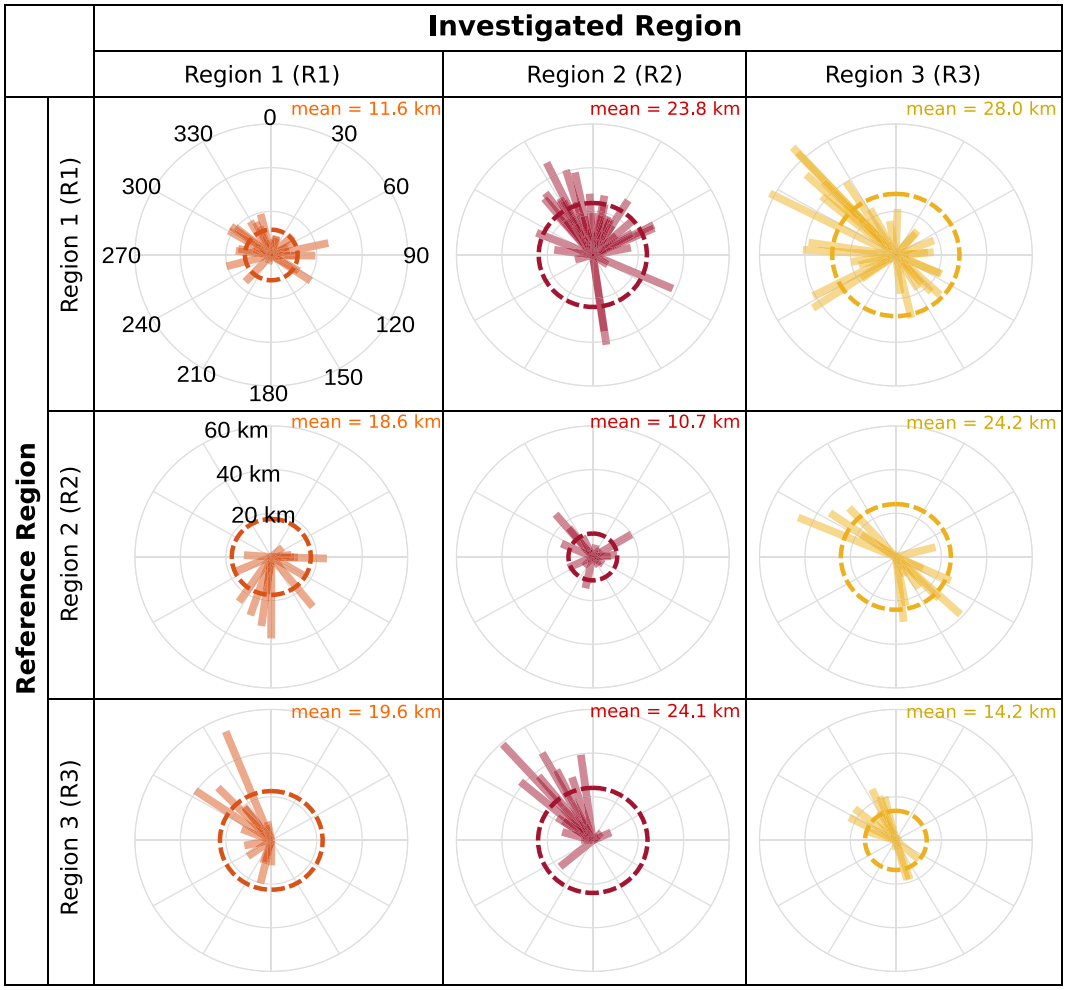

Figure 4. Patterns (direction and amplitude) of spatial biases in the Commander Island region. The reference region represents the region where a reference event is located. The hypocenter correction of a reference event is then applied to all the events in another region (i.e., investigated region) to obtain their Back-projection inferred (BPI) locations. De (both distance and azimuth) of the BPI locations are plotted as bars in the polar plot. In each plot, the mean values of De are marked (thick dashed lines).

(R2→R2/R3→R3) is 12.2 km, which is ~50% smaller than 23.1 km of R2→R1 or R3→R1 (R1→R2 or R3→R2/R1→R3 or R2→R3). In terms of directivities of *De*, for example, we found that R1→R2 is dominated by De offsetting northwards; R1→R3 is dominated by *De* offsetting toward the northwest; R2→R1 is dominated by *De* offsetting toward the south; and both R3→R1 and R3→R2 are dominated by *De* offsetting toward the northwest. Such regionalized features of *De* may be caused by segmented faulting and velocity structure beneath source regions (Figure 1). This finding is somewhat confirmed by Ruppert et al. (2012) who found different zones of seismotectonic activities based on distributions of focal mechanisms in the Commander Island region. In the northwest of our investigated region near Bering Island (169°–170º E, R1 in Figure 1), right-lateral strike-slip displacements are predominant with strike angles subparallel to the island chain (Newberry et al., 1986). A sudden stress change occurred in the southeast of Mednyi Island around 170° E (170°–171º E, R2 in Figure 1), where diverse focal mechanisms exist, including strike-slip events with strike angles subparallel to the elongated direction of Mednyi Island and reverse events whose strike angles are various. Then in the east of 171° E (171°–172° E, R3 in Figure 1), the focal mechanisms experience another change, where the strike angles of reverse events become predominantly between northeast and east (Lutikov et al., 2019). As a result, such structural complexities may be associated with regionalized seismic structures beneath the source region resulting in different patterns of travel-time variations and *De*.

We found that the average amplitude of *De* in the Commander Island region is 21 km and there is a positive correlation between amplitudes of *De* and event separation distances (distances between reference and investigated events).

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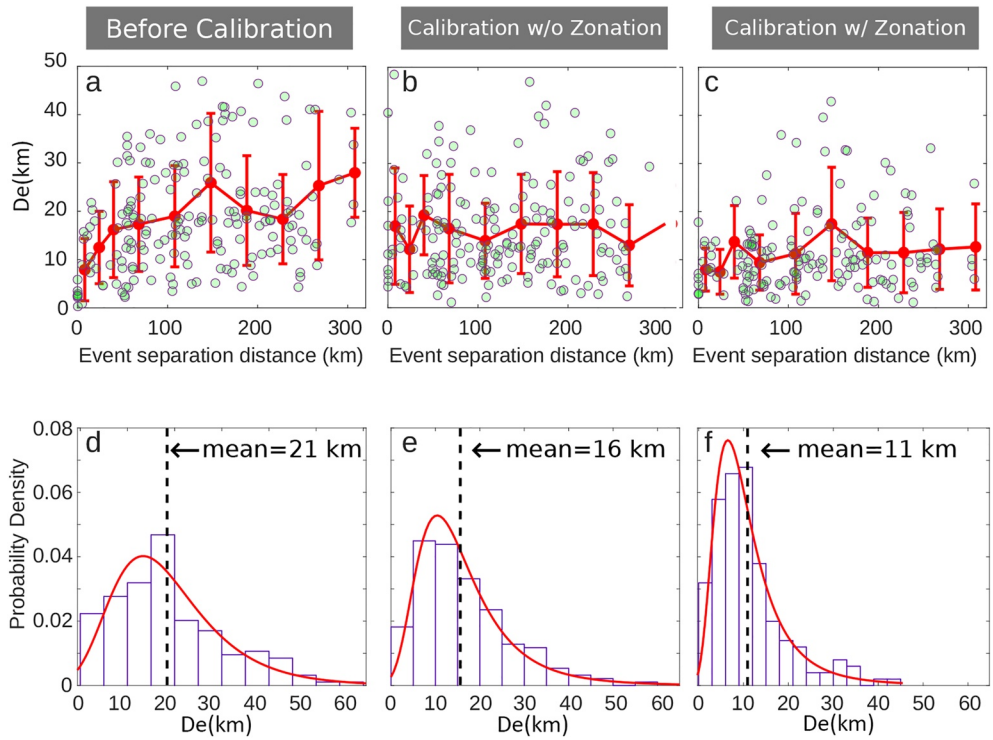

Figure 5. Characteristics of spatial bias before and after slowness calibration. (a–c) De as a function of event separation distances (distances between reference and investigated events), before calibration, after calibration without (w/o) zonation, and after calibration with (w/) zonation. Each data point is one event pair measurement of *De* of the investigated event when applying the hypocenter correction of a reference event. Red error bars are one standard deviation of the measurements. (d–f) Distribution of *De*. The fitted probability density functions (PDF) follow generalized extreme value (GEV) distribution, whose parameters are listed in Table S3 in Supporting Information S1.

Figure 5a shows the 198 measurements plotted as amplitudes of *De* versus event separation distances. In the immediate proximity of a reference event (<150 km), the median *De* increases almost linearly with the separation distance. This is somewhat expected for a slowness correction: if the reference and investigated events share a similar regional velocity structure different from the standard 1D velocity model adopted in BP, there exists a constant slowness error, and the travel-time errors and hence *De* should be proportional to the event separation distance projected onto the direction of the slowness error. Since the orientation of the event separation varies for different reference and investigated event pairs, *De* is loosely, rather than strictly, proportional to the event separation distance. On the other hand, Figure 5a shows that when the event separation is greater than 150 km, *De* seems to plateau and ceases to increase with distance. Such observations indicate that at a larger scale, the difference of velocity structures beneath the reference and investigated events is no longer negligible, and a consistent slowness error does not explain the stochastic variation of travel-time errors and *De*. The fact that *De* plateaus with distance suggests that the travel-time errors are capped by the maximum lateral velocity deviation from the standard 1D Earth model. To further understand the statistics of *De*, we modeled its probability distribution by fitting a generalized extreme value (GEV) distribution (Fan & Shearer, 2017; McFadden, 1978) which measures the probability distribution of a large set of independent, identically distributed random values representing observations. The modeled GEV distribution can be used to more accurately estimate the means, medians, and standard deviations of *De* (Figure 5d, Table S3 in Supporting Information S1). Similar spatial patterns of *De* can also be found in Figure S4a-d and Figure S5a-d in Supporting Information S1 by applying two different BP methods (BMFM and MUSIC BP) in the two frequency bands (0.05–0.5 Hz and 1–4 Hz).

3. The Regional SEBP

3.1. Method

In light of the regional patterns of *De*, we propose to modify the Slowness-Enhanced Back-projection (Meng et al., 2016). Our strategy is to divide a large source region, in this case, a region longer than 400 km, into

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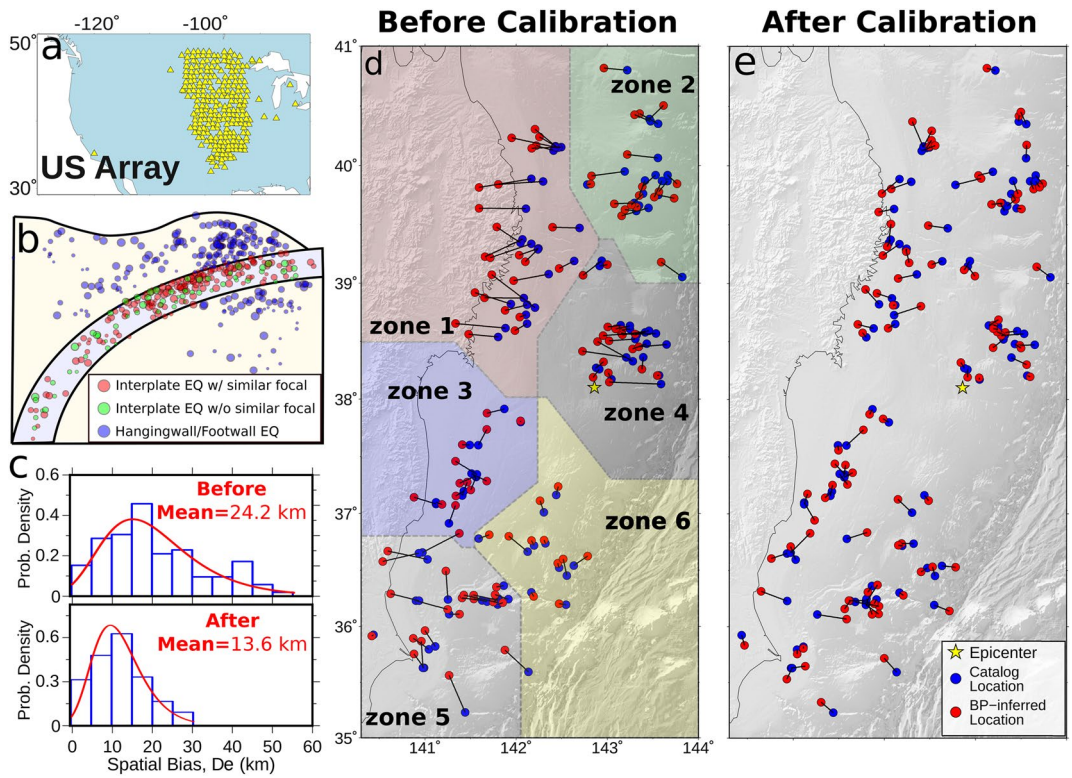

Figure 6. (US array) The effectiveness of the slowness calibration of Back-projection. (a) The US array (267 stations) used in this study. (b) Illustration of the classification of seismic events in the subduction zone based on their relative distance to the slab interface and focal mechanisms. The group of interplate earthquakes with similar focal mechanisms (red circles) as that of the 2011 Mw 9.1 Tohoku earthquake is investigated here. The classification follows Asano et al. (2011). (c) Distribution of *De* of the 109 M4–M7 investigated interplate events with similar focal mechanisms as that of the 2011 Mw 9.1 Tohoku earthquake. The fitted probability density function (PDF) follows a generalized extreme value (GEV) distribution. (d) The spatial biases (black line) of the 109 events, as measured by *De*, the distance of the image peak (BPI location, red circle) from the epicenter (Japan Meteorological Agency (JMA) location with the location uncertainty less than 3 km (Yano et al., 2017, blue circle), which is defined in the same way as Fan and Shearer (2017). Colored regions are segmented automatically by K-means clustering based on the distribution of events' horizontal locations. (e) Same as (d) but after regional SEBP.

sub-regions. Within each sub-region, a uniform slowness correction is derived and applied. We describe our methodology in detail below.

In standard BP analysis, the delay and sum of coherent seismic phases are used to obtain BP images:

$$\mathrm{BP}(\xi, \mathrm{t}) = \sum_j u_j\left(t + T_j^{mod}(\xi) + \delta T_j(\xi)\right) \tag{1}$$

where $\mathrm{BP}(\xi, \mathrm{t})$ is the beamforming power at a potential source location ξ as a function of time t, which can be expressed as a sum over n seismograms. u_j is the normalized seismogram at the jth station, $T_j^{mod}(\xi)$ is the theoretical travel time from the source location ξ to the jth station derived from a reference model, and $\delta T_j(\xi)$ is the travel time error that accounts for the difference between the assumed Earth model and the real Earth velocity structure, which represents the changes in the wavefront curvature and incident angle at the receiver side (Figure 9b). The stack $BP(\xi, t)$ estimates seismic energy released at each time t for all potential source grids, and the location ξ that yields the highest coherence and highest stacked power will be marked as the most likely source location at time t. For large earthquakes, by combining $BP(\xi, t)$ from the whole source period, one can obtain the spatial-temporal process of the earthquake in seismic radiation.

Following eq. (1), the routinely performed "hypocenter correction" measures $\delta T_j(\xi_h)$ by aligning the initial phase with cross-correlation assuming that the first arrival originates from the hypocenter h. Most BP studies assumed that $\delta T_j(\xi_h)$ is applicable for the whole source region, that is,

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$$\delta T_j(\xi) \approx \delta T_j(\xi_h) = T_j(\xi_h) - T_j^{mod}(\xi_h) \tag{2}$$

However, Equation (2) is only exact in the vicinity of the hypocenter, which is why spatial biases exist. Previous applications of slowness calibrations show that spatial biases in one source region follow certain patterns: gradual changes of deviation amplitudes with similar deviation directions (Bao et al., 2019; Meng et al., 2019; Xie et al., 2021). Therefore, we can expand Equation (2) to account for the distance away from the hypocenter $\xi - \xi_h$ using the first-order Taylor expansion:

$$\delta T_j(\xi) = \delta T_j(\xi_h) + \delta\left(\frac{\partial T}{\partial \xi}\right)_j |_{\xi_h} \cdot (\xi - \xi_h) = \delta T_j(\xi_h) + \delta s_j \cdot (\xi - \xi_h) \tag{3}$$

We define $\frac{\partial T}{\partial \xi}$ as δs, the spatial derivatives of travel time errors at the source area, which shares the same unit of [s/km] as slowness. The product of δs and $(\xi - \xi_h)$ can be considered as an inner product of two vectors with respect to the three components (latitude, longitude, and depth):

$$\delta T_j(\xi) - \delta T_j(\xi_h) = \delta s_j^{NS} \cdot (\xi - \xi_h)^{NS} + \delta s_j^{EW} \cdot (\xi - \xi_h)^{EW} + \delta s_j^{UD} \cdot (\xi - \xi_h)^{UD} \tag{4}$$

where EW, NS, and UD denote three orthogonal directions of East-West, North-South, and Up-Down, respectively. Because the teleseismic travel time is not sensitive to the focal depth, we can simplify the equation by ignoring the "vertical" component (UD):

$$\delta T_j(\xi) - \delta T_j(\xi_h) = \delta s_j^{NS} \cdot (\xi - \xi_h)^{NS} + \delta s_j^{EW} \cdot (\xi - \xi_h)^{EW} \tag{5}$$

Since cross-correlation alignment provides the travel time correction relative to the reference seismogram, it will introduce a constant time shift with respect to the reference station. Usually, the choice of the reference station can be arbitrary and would not make any difference because the differential travel time is self-consistent in cross-correlation. To eliminate such a constant time shift, we replace the travel time error $\delta T_j(\xi)$ with the relative travel time error $\delta \hat{T}_j(\xi)$:

$$\delta \hat{T}_j(\xi) = \delta T_j(\xi) - \delta T_{j_r}(\xi) \tag{6}$$

where $\delta T_{j_r}(\xi)$ is the travel time error of the reference station j_r, and Equation (5) can be re-written as:

$$\delta \hat{T}_j(\xi) - \delta \hat{T}_j(\xi_h) = \delta \hat{s}_j^{NS} \cdot (\xi - \xi_h)^{NS} + \delta \hat{s}_j^{EW} \cdot (\xi - \xi_h)^{EW} \tag{7}$$

Now, consider that we have a group of aftershocks (i = 1,2, …, M), and their catalog locations are $\xi_i = \xi_1, \xi_2, \cdots, \xi_i, \cdots \xi_M$. By applying the hypocenter correction, we can obtain their BPI locations ξ_i^{BP}. Considering ξ_i^{BP} as the equivalent of ξ_i in the reference velocity model, we have

$$T_j(\xi_i) - T_j(\xi_h) = T_j^{mod}\left(\xi_i^{BP}\right) - T_j^{mod}(\xi_h) \tag{8}$$

This approximation is supported by the fact that the difference in the two differential travel times between an aftershock and the mainshock hypocenter is an infinitesimal of a higher order than the differential travel time itself. Then, the left-hand side of Equation (6) can be expanded as

$$\delta \hat{T}_j(\xi_i) - \delta \hat{T}_j(\xi_h) = T_j(\xi_i) - T_j^{mod}(\xi_i) - T_{j_r}(\xi_i) + T_{j_r}^{mod}(\xi_i) - T_j(\xi_h) + T_j^{mod}(\xi_h) + T_{j_r}(\xi_h) - T_{j_r}^{mod}(\xi_h) \tag{9}$$

Inserting Equation (8) into Equation (9), we have

$$\delta \hat{T}_j(\xi_i) - \delta \hat{T}_j(\xi_h) \approx \hat{T}_j^{mod}\left(\xi_i^{BP}\right) - \hat{T}_j^{mod}(\xi_i) \tag{10}$$

where

$$\hat{T}_j^{mod}\left(\xi_i^{BP}\right) = T_j^{mod}\left(\xi_i^{BP}\right) - T_{j_r}^{mod}\left(\xi_i^{BP}\right)$$

$$\hat{T}_j^{mod}(\xi_i) = T_j^{mod}(\xi_i) - T_{j_r}^{mod}(\xi_i) \tag{11}$$

Validation of Equation (10) can be found in Text S1 and Figure S6 in Supporting Information S1.

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Combining Equation (7) and Equation (10), we have

$$\hat{T}_j^{mod}\left(\xi_{\mathrm{i}}^{\mathrm{BP}}\right) - \hat{T}_j^{mod}(\xi_h) \approx \delta\hat{s}_j^{NS} \cdot (\xi_i - \xi_h)^{NS} + \delta\hat{s}_j^{EW} \cdot (\xi_i - \xi_h)^{EW} \tag{12}$$

Now, the remaining unknowns, the slowness error vector ($\delta\hat{s}_j^{NS}$; $\delta\hat{s}_j^{EW}$) can be solved by a least-square inversion based on the time-shifts to align the aftershocks. These time-shifts can be obtained in the same fashion as the hypocenter correction. In practice, it is challenging to align the relatively small events (M4–M5.5) because of their low SNRs. We therefore use the differential travel-time of the BPI and catalog locations of events to infer the aftershock timeshifts.

However, for source regions as large as the Commander Island region and the Tohoku region, as we will see in the Results, spatial biases in different sub-regions have different patterns in the amplitudes and directions. This indicates that a uniform slowness calibration over the entire source region cannot effectively account for the spatial biases (Fan & Shearer, 2017; Palo et al., 2014). Therefore, we propose a regional SEBP approach that utilizes different slowness correction terms for each sub-region. Each sub-region k can be represented by its geometrical center ξ^k ($k = 1, 2, ...$) and has its center correction $\delta\hat{T}_j\left(\xi^k\right)$. Following Equation (7), for aftershocks ξ_i located in the region k, we have

$$\delta\hat{T}_j(\xi_i) - \delta\hat{T}_j\left(\xi^k\right) = \left(\delta\hat{s}_j^k\right)^{NS} \cdot \left(\xi_i - \xi^k\right)^{NS} + \left(\delta\hat{s}_j^k\right)^{EW} \cdot \left(\xi_i - \xi^k\right)^{EW} \tag{13}$$

which is equivalent to

$$\delta\hat{T}_j(\xi_i) - \delta\hat{T}_j(\xi_h) = \Delta\delta\hat{T}_j\left(\xi^k\right) + \left(\delta\hat{s}_j^k\right)^{NS} \cdot \left(\xi_i - \xi^k\right)^{NS} + \left(\delta\hat{s}_j^k\right)^{EW} \cdot \left(\xi_i - \xi^k\right)^{EW} \tag{14}$$

where

$$\Delta\delta\hat{T}_j\left(\xi^k\right) = \delta\hat{T}_j\left(\xi^k\right) - \delta\hat{T}_j(\xi_h) \tag{15}$$

denotes the difference between a geometrical center correction and the hypocenter correction.

Using Equation (10), we can rewrite Equation (14):

$$\hat{T}_j^{mod}\left(\xi_{\mathrm{i}}^{\mathrm{BP}}\right) - \hat{T}_j^{mod}(\xi_i) = \Delta\delta\hat{T}_j\left(\xi^k\right) + \left(\delta\hat{s}_j^k\right)^{NS} \cdot \left(\xi_i - \xi^k\right)^{NS} + \left(\delta\hat{s}_j^k\right)^{EW} \cdot \left(\xi_i - \xi^k\right)^{EW} \tag{16}$$

Similar to Equation (12), the least-square inversion can solve for $\left[\Delta\delta\hat{T}_j\left(\xi^k\right), \left(\delta\hat{s}_j^k\right)^{NS}, \left(\delta\hat{s}_j^k\right)^{EW}\right]$. Then, the slowness correction can be mapped to obtain travel time corrections for all source grids. We demonstrate the effectiveness of SEBP in the following section, which is quantified by reductions of spatial biases.

3.2. The Effectiveness of the Regional SEBP

We found that the regional SEBP effectively mitigates the spatial bias in the Commander Island region. Figures 5b and 5c show the amplitudes of *De* versus event separation distances after the uniform SEBP or regional SEBP is applied, respectively. We found that the approximately linear relationship between the median *De* and the separation distance does not exist anymore after the calibration. Figures 5e and 5f show the modeled GEV distribution of De after the uniform SEBP and regional SEBP are applied, respectively. We fit the GEV distribution to obtain location parameters (μ), scale parameters (σ), and shape parameters (κ), which help us to better estimate the means, medians, and standard deviations of De (Table S3 in Supporting Information S1). The location parameter reflects the position of the most frequent observation, while the scale parameter related to the kurtosis of observations reflects the sharpness of the distribution. Table S3 in Supporting Information S1 shows a reduction of the location parameter (μ) of GEV distribution by nearly 30% and 50% for the uniform SEBP and regional SEBP, respectively, compared to μ of the BP without calibration. The scale parameter (σ) is reduced by about 30% and 50% for the uniform SEBP and regional SEBP, respectively. These statistics suggest that when applying the regional SEBP, the GEV distributions of De became narrower and shifted toward the lower values (Figures 5e and 5f). We also found that the mean value of De for regional SEBP reduced from 21 to 11 km, a 48% reduction, which is more significant than that of the uniform SEBP (24% reduction from 21 to 16 km). Besides, Figures S7 and S8 in Supporting Information S1 show patterns of directions and amplitudes of De when the uniform SEBP and regional SEBP are applied, respectively. For any pair of a reference and investigated region, while the directions of De are mostly similar, the average amplitudes of De for regional SEBP are shorter than that of the uniform SEBP.

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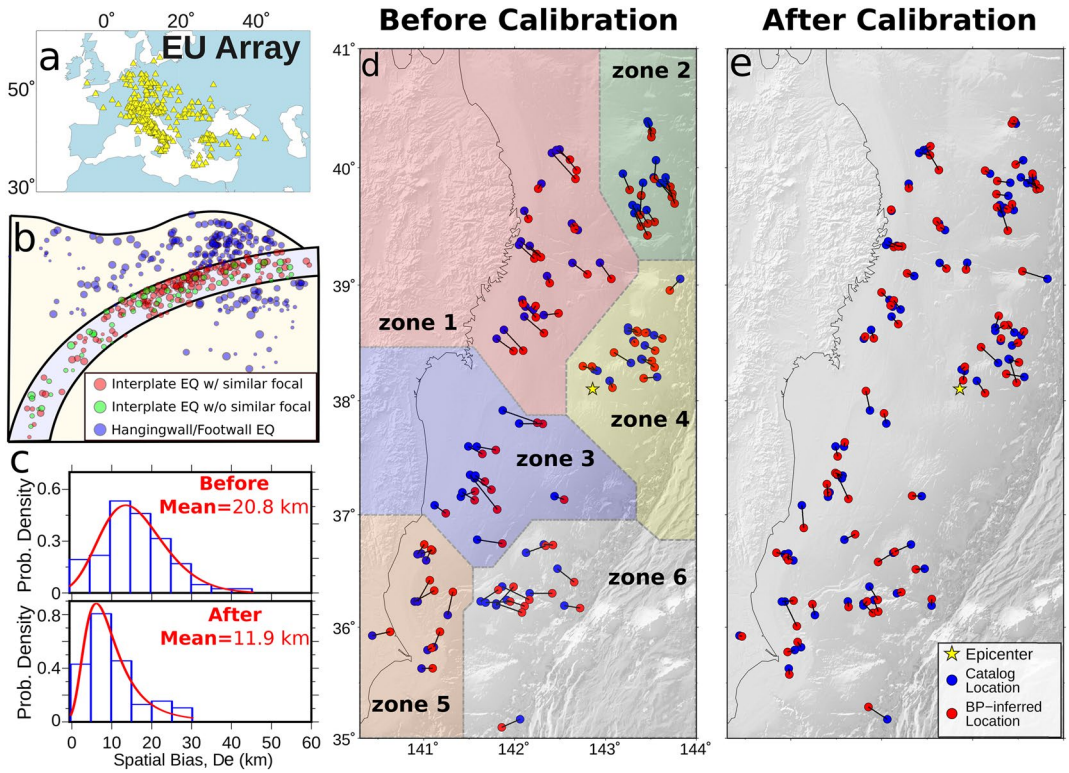

Figure 7. (EU array) The effectiveness of the slowness calibration of Back-projection. (a) The EU array (369 stations) used in this study. (b) Illustration of the classification of seismic events in the subduction zone based on their relative distances to the slab interface and focal mechanisms. The group of interplate earthquakes with similar focal mechanisms (red circles) as that of the 2011 Mw 9.1 Tohoku earthquake is investigated here. The classification follows Asano et al. (2011). (c) Distribution of De of the 82 M4–M7 investigated interplate events with similar focal mechanisms as that of the 2011 Mw 9.1 Tohoku earthquake. The fitted probability density function (PDF) follows a generalized extreme value (GEV) distribution. (d) The spatial biases (black lines) of the 82 events, as measured by De, the distance of the image peak (BPI location, red circle) from the epicenter (JMA location, blue circle). Colored regions are segmented automatically by K-means clustering based on the distribution of events' horizontal locations. (e) Same as (d) but after slowness calibration.

4. Application of SEBP to the Tohoku Region

4.1. Spatial Biases in the Tohoku Region

We further applied the regional SEBP to the source region of the 2011 Mw 9.1 Tohoku earthquake, one of the largest and most complicated earthquakes ever recorded. Seismic velocity structure in subduction zones contains strong contrast: high P-wave velocities inside the cold slab and low P-wave velocity in the mantle wedge. Therefore, the velocity structures surrounding interplate and intraplate events may be significantly different even if they are close by. For our analysis, we choose only the interplate events (Figure 6b), not intraplate events in the hanging wall or footwall, to make sure that ray paths go through similar velocity structures. This procedure also ensures that the derived slowness corrections are applicable to the mainshock which occurs primarily along the plate interface. The selection criteria of interplate events follow that of Asano et al. (2011): (a) rake angle $\lambda \geq 0°$, (b) Kagan's angle (Kagan, 1991) $\theta \leq 35°$, (c) The difference between the centroid depth and the plate boundary $\delta d \leq 20$ km. Kagan's angle is the minimum rotation angle of the focal mechanisms relative to a reference mechanism, which is used to select events with similar focal mechanisms as the megathrust mainshock. The reference mechanism of the 2011 Mw 9.1 Tohoku earthquake is set as strike $\varphi = 195°$, dip $\delta = 13°$, rake $\lambda = 90°$ (USGS). The Slab 2.0 model (Hayes et al., 2018) is used to constrain the depths of the slab interface.

A total number of 109 M4–M7 interplate events in the Tohoku region are investigated (Figure 6). The extensive spatial coverage of earthquakes in this region provides us with an unprecedented number of samples to study the *De* of BP. We used both the USarray (US, 267 stations) and European array (EU, 369 stations) so that the results can be cross-examined. Waveform data are downloaded from the Data Management Center (DMC) of

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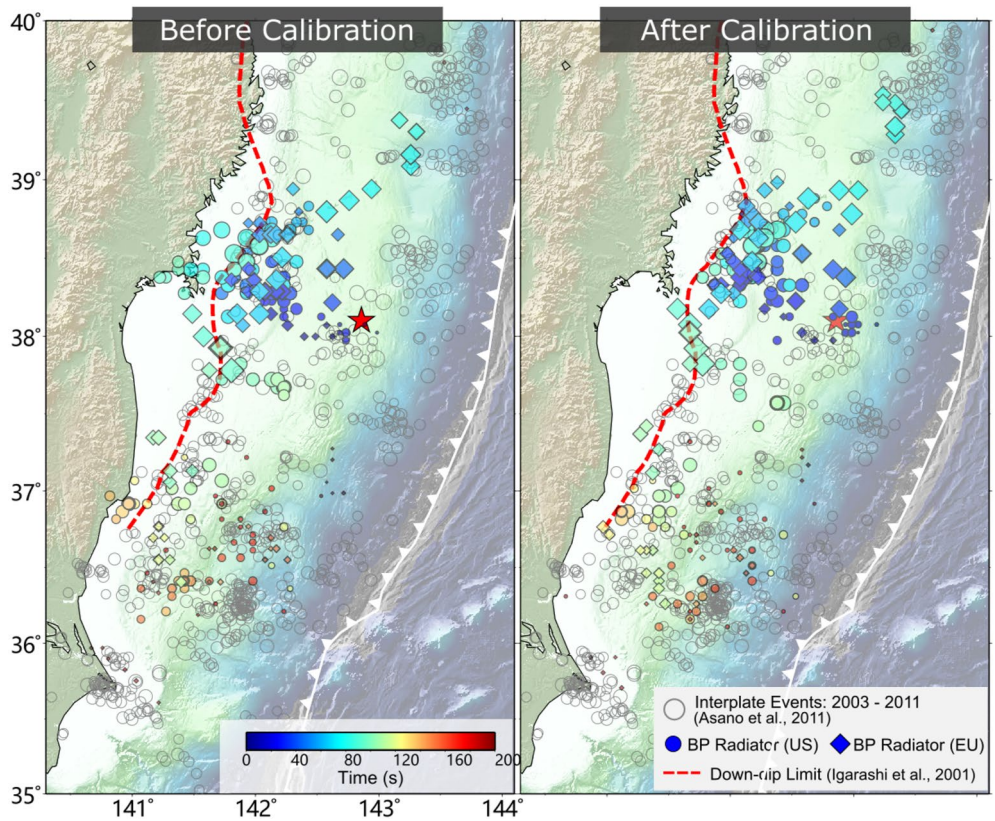

Figure 8. Coseismic rupture evolution of high-frequency (0.5–2 Hz) radiations before (left) and after (right) slowness calibration. BPs are imaged by the US array (circle) and EU array (diamond), with sizes proportional to the relative energy and color representing the rupture time. The red dashed line shows the down-dip limit of interplate seismicity (Igarashi et al., 2001).

the Incorporated Research Institutions for Seismology (IRIS) (https://ds.iris.edu/wilber3) and Observatories & Research Facilities for European Seismology ((ORFEUS) (http://orfeus-eu.org/)).

Figures 6d and S9 in Supporting Information S1 show the spatial biases *De* of the 109 investigated events with the Tohoku mainshock as the reference event. Clear patterns of *De* are found in some regions, for example, in the northwest (offsetting westward) and southwest (offsetting northwestward), while the patterns in other regions are less systematic. We again attempt to reduce *De* using regional SEBP. To properly group the investigated events into regional clusters, we adopt K-means clustering based on the event locations (Forgy, 1965; Macqueen, 1967). It aims to partition N observations into K clusters in which each observation belongs to the cluster with the nearest mean. We are aware that the choice of the value of K (number of clusters) can be subjective so we tried four different values ($K = 4$, 5, 6, and 7). Since $K = 6$ gives the most visually consistent clusters, we consider it optimal and show its results in Figure 6d. The results show that a regional SEBP strategy effectively reduces *De* in the Tohoku region. Figure 6c shows the modeled GEV distribution of *De* before and after calibration, and Figure 6e shows the residue *De* after calibration. SEBP reduced the mean value of *De* from 24.2 to 13.6 km, a 44% reduction. The effectiveness of regional SEBP is also demonstrated with the European array. Similar to Figure 6, Figure 7 shows the spatial biases of the Tohoku region before and after calibration. The mean value of *De* was reduced by 43% from 20.8 to 11.9 km.

4.2. Application of SEBP to the 2011 M9.1 Tohoku Earthquake

By further applying the SEBP to the mainshock of the 2011 M9.1 Tohoku earthquake, we found that the sources of high-frequency radiations of the 2011 Mw 9.1 Tohoku earthquake likely did not expand beyond the down-dip limit of the seismogenic zone. Figure 8 shows the US and EU BP results of the 2011 Tohoku mainshock before and after the slowness calibration. The results are overall consistent with previous BP studies (Ishii, 2011; Koper et al., 2011; Meng et al., 2011; Wang and Mori, 2011; Yao et al., 2011), where high-frequency radiations initiated

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with a slow down-dip expansion and then propagated toward the southwest (see Lay, 2018, for a comprehensive review of the 2011 Tohoku earthquake). Some discrepancy can also be observed between the US and EU BP results after the slowness calibration, the HF radiators from the EU array are significantly stronger than those from the US array in the northern part of the source region. It is because the HF radiators resolved by BP are the centroid locations of beamforming power at each time, the location information of the whole circular rupture front cannot be extracted, which may lead to the discrepancy in rupture directivity for different seismic arrays when the rupture propagates toward more than one direction (Zhang et al., 2022). Besides, the source-receiver geometry and the apertures of the seismic array will also affect the resolution of the BP results. However, what we found differently is that in the uncalibrated BPs, high-frequency radiators went beyond the seismogenic down-dip limit, while those of the calibrated SEBP stopped right above the boundary. The down-dip limit shown in Figure 8 is based on the bottom boundary of the interplate seismicity that occurred from April 1992 to August 1998, which are defined using the same criteria as mentioned above (rake angle, Kagan angle, and the relative distance to the slab interface) (Igarashi et al., 2001). The depth of the down-dip limit ranges from 40 to 50 km estimated from the upper boundary of the subducting Pacific slab (Kita et al., 2010; Nakajima & Hasegawa, 2006).

5. Discussion and Conclusion

We systematically investigated the characteristics of *De* in the Commander Island and Tohoku regions as well as the effectiveness and limitation of the Slowness Enhanced Back-projection (SEBP). We found the average *De* in both regions is ∼20 km, and the amplitude of *De* has a positive correlation with the hypocenter distance. We also found that for the two investigated regions, the regional SEBP is more effective than the uniform SEBP and has reduced *De* by ∼50%, to an average value of ∼10 km. We argue that the requirement of the regional SEBP is likely because the two investigated source regions are large (400–500 km) and the velocity structures beneath the source region cannot be considered uniform. It also suggests that the previous applications of the uniform SEBP (e.g., the 2015 Mw 7.8 Gorkha (Meng et al., 2016), the 2015 Mw 7.2 Tajikistan (Sangha et al., 2017), the 2015 Mw 8.3 Illapel (Meng et al., 2018), and the 2018 Mw 7.5 Palu earthquakes (Bao et al., 2019)) are likely due to their relatively small rupture lengths (≤150 km) and hence smaller lateral velocity variation. We also note that the K-means clustering based on event locations is effective in partitioning zones in a regional SEBP study. Other location-based clustering methods like PAM clustering (Schubert & Zimek, 2019) and OPTICS clustering (Ankerst et al., 1999) may also be effective.

As the earthquakes we selected in the Tohoku region are all interplate earthquakes, the heterogeneity of the velocity structure in the subduction Pacific slab and its subslab region becomes the dominant source of different slownesses along ray paths, resulting in the regionalized patterns of the investigated events' spatial bias. To explore the relationship between the slab and subslab heterogeneity in the Tohoku region and the subregions we automatically derived by K-means clustering based on the spatial bias patterns, we first referred to Liu and Zhao (2016) to compare the slab heterogeneity with reference to the IASP91 Earth model and our derived subregions by plotting subregions onto the average Vp anomaly distribution in the slab. We found that the subregions we statistically segmented the Tohoku region into for both the US and EU arrays are correlated with the slab Vp anomaly distribution. For example, in the northern Tohoku area, zones 1 and 2 appear to match well with the high- and low-Vp anomaly slab regions shown in Figure S12(a-b) in Supporting Information S1. In the southwestern area, zones 3 and 6 correspond well to the high- and low-Vp anomaly slab regions, respectively.

We then explored the relationship between the subslab heterogeneity in the Tohoku region and the subregions. With reference to the IASP91 Earth model, Fan and Zhao (2021) derived the high-resolution tomographic P-wave velocity (Vp) anomalies in the Tohoku region using P-wave arrival times of local and teleseismic events recorded at seismic stations in and around the Japan Islands. Based on that, we obtained the subslab Vp anomalies in this region averaged from the lower boundary of the Pacific slab to the depth of ∼310 km, and then plotted subregions we derived for the US and EU arrays onto the tomographic model, as shown in the figure below. We found that the high- and low-Vp anomaly regions in the subslab region are basically consistent with those regions in the slab. Similarly, we also found that the subregions we statistically segmented the Tohoku region into for both the US and EU arrays are correlated with the subslab Vp anomaly distribution. For example, in the northern Tohoku area, zones 1 and 2 appear to match well with the high- and low-Vp anomaly subslab regions shown in Figure S12(c-d) in Supporting Information S1. In the southwestern area, zones 5 and 6 correspond well to the low-Vp anomaly subslab regions, respectively.

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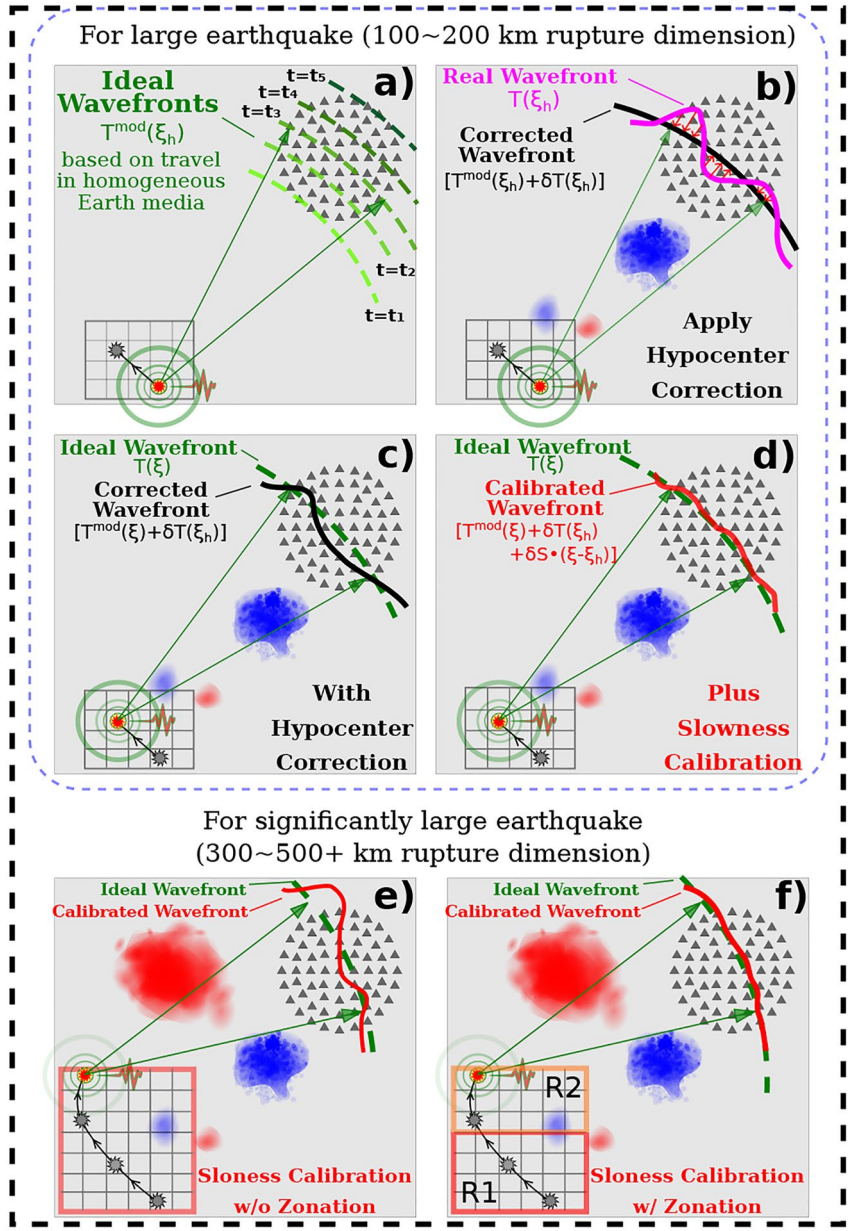

Figure 9. Cartoon illustration of the Back-projection method and slowness calibration. (a) Geometries of wavefronts (green dashed lines) depicted by the travel time table from a source region to a seismic array. The wavefronts are ideally smooth spherical if the Earth media is homogeneous. (b) The structural heterogeneity of the real Earth perturbs travel times and results in a distorted wavefront (wiggly pink line). By aligning P wave arrivals with cross-correlation, relative travel time errors of the ray paths originating from the hypocenter can be well corrected (i.e., hypocenter correction). (c) However, the hypocenter corrections are only exact at the hypocenter, because rays originating from other sources will travel through different 3-D structures, which results in a distorted wavefront (wiggly black line) again. (d) The slowness calibration effectively mitigates the waveform distortion as spatial biases in the entire source region are accounted for, which resulted in less distorted wavefronts (red solid line). (e) Such a uniform slowness calibration becomes less effective when the source region is significantly large (e.g., 300+ km), and calibrated wavefronts (red wiggly line) that originate from sources far away from the hypocenter are not effectively corrected. This is likely caused by the fact that rays from sources that are far away travel through different large-scale heterogeneities. (f) When applying the regional SEBP, travel-time errors in the entire source region can be better calibrated and result in less distorted wavefronts (red solid line).

Zeng et al. (2022) also address the spatial bias in BPs caused by the source-side 3D velocity variation by accounting for travel-time perturbations (ΔT) through resolving the first-order spatial derivatives of ΔT along the ray path, which is equivalent to the slowness error—δs, defined in the original SEBP (Meng et al., 2016), divided by

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the known theoretical slowness s^t. Their method and the SEBP are different in how to derive the slowness error. SEBP first derives the BPI locations of the calibration events by adopting the reference event's hypocenter corrections and then resolves the slowness error by calibrating the BPI locations of calibration events into their regional or local relocated locations, using the least-square regression approach. The advantage of our method is that we do not need to align the waveforms of the calibration events, which can be challenging for low-magnitudes (∼5) events with low SNRs recorded at teleseismic distances. While Zeng et al. (2022) first group calibration events into 3-event groups and then estimate the slowness error along the ray path through a simple matrix product of the travel-time error differences (See Equation 5 in their paper). In that sense, direct estimates of the travel time error are derived by cross-correlating waveforms of calibration events rather than being inferred from BPI locations in our method. However, such estimates based on waveform cross-correlation require high-quality teleseismic waveforms of calibration events, which may result in fewer available calibration events than our method. Under such conditions, they claim that the 3-event cluster strategy is simple and may achieve a good spatial resolution of slowness error even when calibration events are scarce. So, future work will be needed to make a quantitative comparison of the SEBP and their method on the performance of mitigating spatial bias of BP.

5.1. Travel-Time Corrections of SEBP

The SEBP tackled the problem of spatial bias by mitigating the leading source of errors—travel time errors due to the 3-D Earth structure. Figure 9a shows an ideal scenario where wavefronts emitted from an event hypocenter T^mod (ξ_h) at teleseismic distances are smoothly spherical and can be predicted under a 1D reference Earth model. In reality, the regional structure of the real Earth (blue-shaded area) perturbs travel times and results in distorted wavefront T(ξ_h) (Figure 9b). By aligning the first P wave arrivals, relative time errors $\delta T(\xi_h)$ of the ray paths originating from the hypocenter can be calibrated (i.e., hypocenter correction). However, as illustrated by Figure 9c, the hypocenter corrections are only exact at the hypocenter. As the rupture front moves away within a certain distance range, the rays originating from the assumed rupture can be still considered to travel through a similar regional structure as that under the hypocenter. But the heterogeneity of the regional structure induces the spatial gradient of the travel-time (slowness), which also needs to be corrected by the slowness calibration δs to restore smooth wavefronts (Figure 9d). Our study shows that if the rupture length is less than 150 km, a uniform slowness in the entire source region is sufficient to calibrate the travel-time errors. However, when the source region is significantly larger like the two investigated regions in our study, a uniform slowness calibration becomes less effective, because the rays from subsequent sources far away from the hypocenter travel through a different regional structure (red-shaded area) (Figure 9e). Figure 9f illustrates that by allowing different slowness terms in each region, the calibration can more effectively correct travel time errors and mitigate spatial biases. We remark that by implementing all the travel-time corrections, the regional SEBP reduces *De* by ∼50%. This indicates that half of *De* are aleatory uncertainties that are caused by regional structural complexity and can be calibrated with aftershocks, while the residue errors are epistemic uncertainties caused by local velocity heterogeneities that are unique for each aftershock.

5.2. Down-Dip Penetration of the 2011 Tohoku Earthquake

Our SEBP result of the 2011 Tohoku earthquake shows that its coseismic slip likely did not penetrate the seismogenic down-dip limit, which may provide unique insights into the rupture dynamics of megathrust earthquakes. Results of previous observational studies indicate that the 2011 Tohoku earthquake may have had a down-dip penetration. For example, previous BP studies show that sources of high-frequency radiations extended below the down-dip limit which is shown in Fig. S10 in Supporting Information S1 (Koper et al., 2011; Meng et al., 2011; Yao et al., 2011), and many finite-fault models indicate significant coseismic slip below the down-dip limit: 10–20 m (Ide et al., 2011), 5–10 m (Ammon et al., 2011; Melgar et al., 2013; Yue & Lay, 2011), and 4–8 m (Hayes, 2011). Most of these studies did not focus on this feature and did not provide any discussion. However, if such a down-dip penetration is true, it should be of great interest because it challenges the traditional view that ruptures stop at the brittle-ductile transition. In a standard conceptual model for active faults (Figure 10), seismogenic (brittle) zones are mainly controlled by velocity-weakening (VW) frictions which host large earthquakes with fast seismic slips, while the ductile regions, located beyond and below the seismogenic zone, are often associated with velocity-strengthening (VS) frictions which only host aseismic creeping and slow earthquakes (Burgmann et al., 2000; Chlieh et al., 2008; Perfettini et al., 2010). Therefore, it would be surprising if the coseis-

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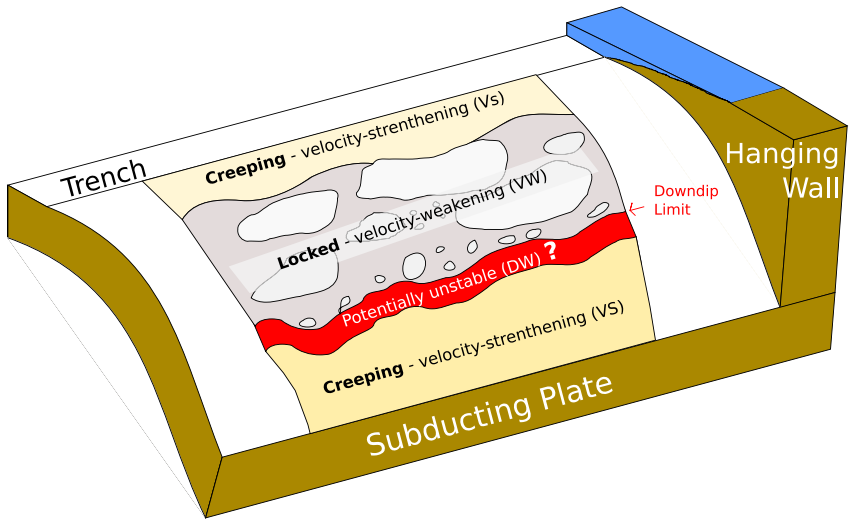

Figure 10. A conceptual model for the interplate megathrust fault. The seismic locked zones are controlled by velocity-weakening (VW) frictions while the creeping ductile zones are associated with velocity-strengthening (VS) frictions. Dramatic weakening (DW) activated at a high slip rate is one candidate mechanism to explain deep penetration beyond the down-dip limit of interplate seismicity.

mic slip of the 2011 Tohoku earthquake penetrated the ductile zone. We elaborate this matter further here given the remarkable agreement between the SEBPs of both the US and EU arrays: the sources of high-frequency radiations did not expand beyond the down-dip seismogenic limit. Moreover, Figure S11 in Supporting Information S1 provides another example showing that the high-frequency radiators of the 2015 Mw 8.3 Illapel earthquake stopped right at the down-dip limit of interplate seismicity (Meng et al., 2018). Jiang and Lapusta (2016) investigated the possible mechanisms of down-dip penetrations on strike-slip faults using dynamic rupture simulations. They found that activation of dramatic weakening (DW) due to thermal pressurization of pore fluid enables rupture into the otherwise aseismic velocity strengthening region (Figure 10). Such an enhanced dynamic weakening of fault frictional resistance occurs at slip rates $\geq$ 0.1 m/s (Di Toro et al., 2011; Rice, 2006), which has critical impacts on earthquake rupture propagation and effectively turns the creeping fault regions into seismic ones (Noda & Lapusta, 2013; Platt et al., 2014). However, the absence of deep penetration beyond the down-dip seismic limit in the Tohoku and Illapel earthquake suggests that the coseismic slip rate at the bottom of the seismogenic zone was not fast enough to trigger the dramatic weakening (DW) at the ductile region of the megathrust. An alternative scenario is that the down-dip penetration did occur but the BP is incapable of resolving it. This is possible if the rheology in the ductile region is too smooth and lacks material heterogeneity to efficiently generate strong high-frequency radiations. In that sense, a similar phenomenon of little high-frequency radiation at shallow depths during the Tohoku earthquake might also be attributed to the smooth velocity-strengthening region near the trench. We consider this as a less plausible scenario. If such deep penetration did exist, the stress shadow caused by the deep coseismic slip would shut off aftershocks near the pre-seismic down-dip limit. However, this is not the case based on the distribution of interplate aftershocks in Tohoku (Asano et al., 2011). We note that further numerical scrutiny and a systematic review of down-dip rupture extension of subduction zone earthquakes should be warranted.

Data Availability Statement

Our teleseismic broadband waveforms were obtained through the IRIS Wilber3 system (https://ds.iris.edu/wilber3/) or IRIS Web Services (https://service.iris.edu), including the following seismic networks: (a) the CH (Switzerland Seismological Network; Swiss Seismological Service, 1983); (b) the CZ (Czech Regional Seismic Network; Charles University in Prague (Czech), Institute of Geonics, Institute of Geophysics, Academy of Sciences of the Czech Republic, Institute of Physics of the Earth Masaryk University (Czech), & Institute of Rock Structure and Mechanics, 1973); (c) the GB (Great Britain Seismograph Network; British Geological Survey); (d) the GE (GEOFON Seismic Network; GEOFON Data Centre, 1993); (e) the GR (German Regional Seismic Network (GR); BGR Federal Institute for Geosciences and Natural Resources (BGR, Germany), 1976); (f) the

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GU (Regional Seismic Network of North Western Italy (RSNI); University of Genoa (UniGe), 1967); (h) the II (IRIS/IDA Seismic Network; Scripps Institution Of Oceanography, 1986); (i) the IV (Italian National Seismic Network; Istituto Nazionale di Geofisica e Vulcanologia, 2005); (j) the KO (Kandilli Observatory And Earthquake Research Institute (KOERI) (KOERI); Kandilli Observatory And Earthquake Research Institute, Boğaziçi University (KOERI), 1971); (j) the MN (Mediterranean Very Broadband Seismographic Network; Istituto Nazionale di Geofisica e Vulcanologia, 2005); (k) the RO (Romanian Seismic Network; National Institute for Earth Physics, 1994) (g) the TA (USArray Transportable Array; IRIS Transportable Array, 2003); (j) the US (U.S. National Seismic Network; Albuquerque Seismological Laboratory, 1990). The catalog data and arrival time data of earthquakes for the period of 2011–2016 in the Commander Island were provided by the EHB catalog (http://www.isc.ac.uk/ehbbulletin/search/catalogue/, Engdahl et al. (2020); International Seismological Centre (2020)) and from the U.S. Geological Survey Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat, https://earthquake.usgs.gov/data/comcat/) for the time period of 2016–2017. We are grateful to the United States Geological Survey (USGS, 2017; https://earthquake.usgs.gov/earthquakes/map/) and the Global Centroid Moment Tensor Catalog(https://www.globalcmt.org/, Ekström G et al., 2012) for the earthquake catalog with moment tensor solutions.

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